THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 3: **Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The anode mud in refining of copper contains: 1

(a) sodium **(b)** zinc **(c)** carbon only **(d)** silver and gold

2. The reducing agent in the thermit reaction is: 1

(a) hydrogen **(b)** aluminium **(c)** iron oxide **(d)** carbon

3. Which element forms a covalent compound with chlorine? 1

(a) carbon **(b)** sodium **(c)** calcium **(d)** magnesium

4. Mercury is used in thermometers because it is: 1

(a) a solid **(b)** very reactive

(c) liquid and expands uniformly **(d)** a poor conductor

5. In the extraction of sodium from molten NaCl, the product at the cathode is: 1

(a) sodium **(b)** hydrogen **(c)** chlorine **(d)** oxygen

6. The number of electrons lost by aluminium to form its ion is: 1

(a) 2 **(b)** 4 **(c)** 3 **(d)** 1

7. The alloy used for welding electric wires is: 1

(a) bronze **(b)** steel **(c)** solder **(d)** brass

8. Which of the following will give a thin layer of oxide that protects the metal? 1

(a) potassium **(b)** aluminium **(c)** sodium **(d)** iron

9. Aluminium chloride has the formula: 1

(a) $AlCl_3$ **(b)** $AlCl_2$ **(c)** Al_2Cl_3 **(d)** AlCl

10. Which of the following is NOT an alloy? 1

(a) brass **(b)** sodium chloride **(c)** bronze **(d)** steel

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State one use of the thermit reaction and write the equation. 2

12. What happens when copper is heated in air? Write the equation. 2

13. Write the equation of the reaction of zinc oxide with sodium hydroxide. 2

14. What is meant by 24-carat gold? Why is 22-carat gold preferred for jewellery? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why does aqua regia dissolve gold but HCl or HNO_3 alone do not? **3**

16. What are the products when sodium reacts with chlorine? Explain the bond. **3**

17. Explain the role of flux and slag in the extraction of metals. **3**

OR

How is copper obtained from copper glance? Write the equations. **3**

18. Why are metals good conductors? Why is graphite also a conductor? **3**

OR

Explain what happens when an iron nail is dipped in **3**
(i) $CuSO_4$ and
(ii) $ZnSO_4$ solutions.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the formation of ionic compounds MgO and Na_2O, and give four properties. **5**

OR

Explain the properties of an alloy and why pure gold is made into 22-carat gold. Name two alloys of copper. **5**

20. Explain the electrolytic refining of copper using a labelled description. What happens to impurities? **5**

OR

Explain why zinc is amphoteric and why aluminium oxide is protective. Write related equations. **5**

END OF QUESTION PAPER — Total: 40 Marks

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